# EE708 Term Projects

## Project 1: ShiftGuard10: Robust Image Classification

**Dataset Description:** 32×32 RGB **images** from **10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck)**

- Training set: labeled images (train_images/) + labels file (train_labels.csv)
- Test set: unlabeled images (test_images/) + submission template (sample_submission.csv)
- Class names: listed in classes.txt

**Task**: Train a model using only the training data and predict the class label for each test image.
**Metric**: Macro F1 (F1-score with macro averaging across 10 classes)
**Submission for code evaluation:** Submit a Notebook that will generate a CSV with exactly two columns:

- `id`: image id (e.g., 000001)
- `label`: predicted class name (must be one of the 10 class names)

Use sample_submission.csv as the template. Follow its formatting exactly.

## Project 2: Digit Recognition

**Dataset Description:** Short **.wav** audio clips of spoken digits from **10 classes (0–9)**

- Training set: labeled audio clips (**train_audio/**) + labels file (**train.csv**)
- Test set: unlabeled audio clips (**test_audio/**) + test IDs file (**test.csv**)
- Audio format: all files are provided as **.wav**
- File-ID mapping: each CSV row contains an **id** that maps directly to a file (use the id **without** the .wav extension)
    - Training audio path: train_audio/{id}.wav
    - Test audio path: test_audio/{id}.wav

**Task:** Train a model using only the training data to recognize the spoken digit, and predict the digit label for each test audio clip.
Note: Test data contains unseen speakers and noisy/augmented recordings, so avoid overfitting to speaker identity or clean-audio artifacts; aim for digit-relevant acoustic cues that generalize across speaker and recording variations.
**Metric**: Macro F1 (F1-score with macro averaging across 10 classes)
**Submission for code evaluation:** Submit a CSV with exactly two columns:

- `id`: must match entries in test.csv
- `label`: predicted digit class (must be an integer in {0,1,2,3,4,5,6,7,8,9})

Example submission:
id,label
test_000001,4
test_000002,0
test_000003,9

# Rules

1. **Kaggle Team:**
    - After joining the competition, one team member must send an invite to all other group members once they have also joined.
    - ***The team name should be the Group number, for example, "Group 1".***
2. **Notebook-only submissions**
    - You must submit using a Kaggle Notebook (Notebook-based submission).
    - File upload-based submission is not allowed.
    - Do not upload a CSV generated outside Kaggle.
    - Your notebook must:
        - train the model from scratch
        - Run inference on the test set
        - Write the submission CSV
3. **No external data /no pretrained models**
    - You may use only the competition data provided.
    - No external datasets, no extra images, no web data, no scraping.
    - The model must be trained from scratch.
    - No pretrained weights (ImageNet, self-supervised checkpoints, etc.).
    - No importing external checkpoints from Kaggle Datasets.
    - No external data /no pretrained models
4. **Reproducibility required:** Your notebook must be reproducible:
    - include training + inference code
    - run end-to-end on Kaggle
    - avoid hidden manual steps
5. **Integrity/Fair Play**
    - Do not attempt to reverse-engineer hidden labels.
    - Do not copy other teams' outputs.
    - Do not exploit loopholes in the evaluation process. Violations may lead to disqualification.

# Evaluation

| | **Marks** |
|---|---|
| Code + Leaderboard | 12 |
| Presentation | 10 |
| Report | 3 |
| **Total** | **25** |

**Code:** Score will be calculated using 40% of the public and 60% of the private score:

$$final_score = 0.4 \times public_score + 0.6 \times private_score$$

- The public score is computed on the visible test dataset and will be displayed during the competition.
- The private score will be released after the competition concludes and is computed on a private test dataset not accessible to participants.

**Presentation:**

- Each group must prepare slides and deliver a clear end-to-end summary of your work (problem, dataset understanding/EDA, approach, experiments, results, and key takeaways).
- **The number of slides is limited to 10**.
- Every team member must present at least one part of the presentation.
- Q&A requirement: During the question-answer session, all team members are expected to respond to any question related to the project, irrespective of which specific component they worked on. This is to ensure shared ownership and a complete understanding of the full pipeline.
- Each group will be given a maximum of **10 minutes for presentation**. The presentation should be brief and concise.

**Report:**

- The **report is limited to 3 pages**, with an additional page allowed for references.
- The report should be in the **IEEE conference template**, either Microsoft Word or LaTeX. The template can be accessed here: IEEE Template.
- The report must include the objective, preprocessing steps, model architecture, performance of the final model, and an ablation study.
- You should also highlight the individual contributions in the report.
- If plagiarism or data manipulation is found, a penalty will be imposed irrespective of the quality of implementation and presentation.

**Final submission consists of the following files:**

- Code.ipynb
- Report.pdf